

Unit 17

CARBONYL COMPOUNDS

30 items · 72 marks

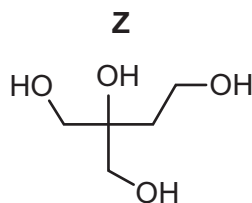
Syllabus sections in this unit

17.1 Aldehydes and ketones

Unit 17: Carbonyl compounds**1. M/J 26/P21/Q4(b)(ii)**

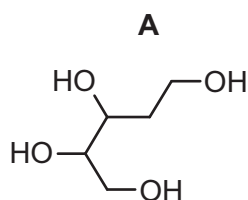
Context: part (a) (introduction)

- 4 (a)** Compound **Z** is oxidised to produce organic compound **W**, $\text{C}_5\text{H}_6\text{O}_7$, by heating under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$.

**Figure 4.1**

Context: part (b) (introduction)

- (b)** Compound **A** is a structural isomer of **Z**.

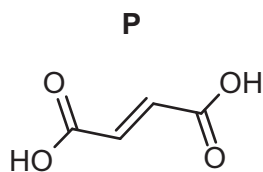
**Figure 4.2**

- (ii)** Suggest why Fehling's reagent does **not** react with **A** but does react with aldehyde functional groups.

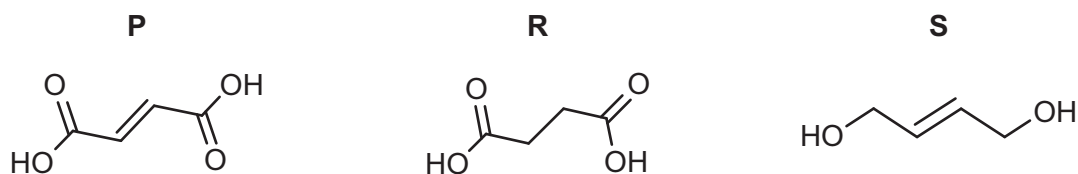
[1]

2. M/J 26/P22/Q6(c)(iii)

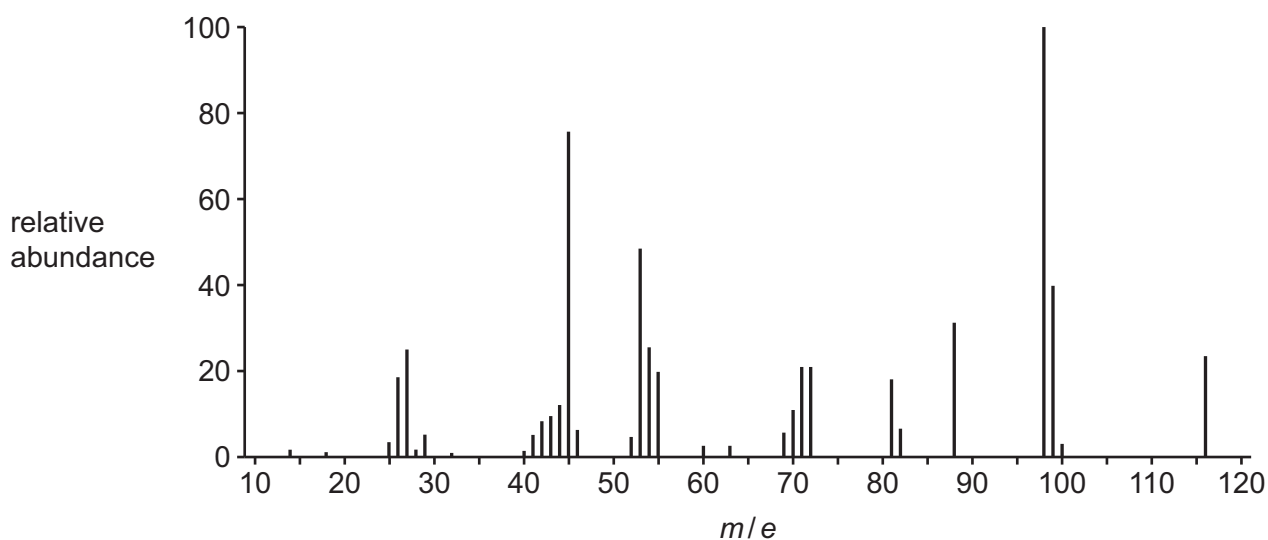
Context: part (a) (introduction)

6 (a) Figure 6.1 shows the structure of compound **P**.**Figure 6.1**

Context: part (c) (introduction)

(c) An unlabelled sample contains one of compounds **P**, **R** or **S**.**Figure 6.4**

The mass spectrum of the sample is shown in Figure 6.5.

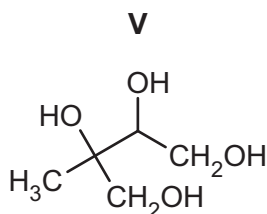
**Figure 6.5****(iii)** Suggest **one** reagent that is used to distinguish between samples of **R** and **S**. Include appropriate observations.

[2]

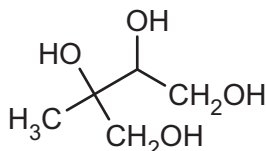
3. M/J 26/P23/Q4(a)(iii)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

**Figure 4.1**

Context: Fig. 4.1

**Figure 4.1**

- (iii) Suggest why Fehling's reagent does **not** react with **V** but does react with aldehyde functional groups.

[1]

4. M/J 25/P21/Q6(a)

Context

- 6** Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.
- (a) State the functional group in ethanal, propanal and propanone.

[1]

5. M/J 25/P21/Q6(b)

Context

- 6** Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.
- (b) State a reagent and the relevant observation that confirm that **F**, **G** and **H** have the same functional group.
- reagent
- observation

[2]

6. M/J 25/P21/Q6(c)(i)-(ii)

Context

- 6** Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.

Context: part (c) (introduction)

- (c)** Separate samples of **F**, **G** and **H** are each tested with Tollens' reagent and with alkaline $I_2(aq)$. The observations are shown in Table 6.1.

Table 6.1

	Tollens' reagent	alkaline $I_2(aq)$
F	no observable change	pale yellow precipitate
G	silver mirror	pale yellow precipitate
H	silver mirror	no precipitate

- (i)** Use Table 6.1 to name the organic compounds in bottles **F**, **G** and **H**.

F =**G** =**H** =

[2]

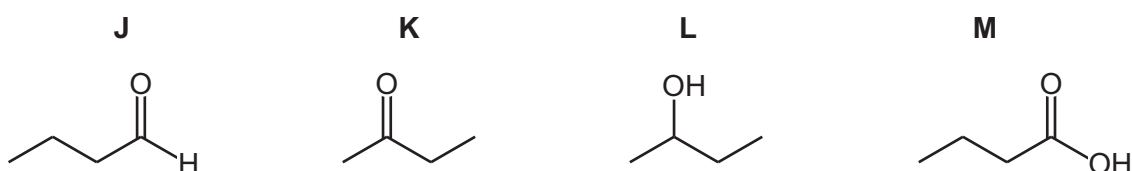
- (ii)** Identify the yellow precipitate produced when alkaline $I_2(aq)$ is added to separate samples of **F** and **G**.

[1]

7. MAR 25/P22/Q4(b)(i)

Context

- 4** Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

**Fig. 4.1**

Context: part (b) (introduction)

- (b)** **K** reacts with HCN in the presence of a KCN catalyst, forming **N**.

- (i)** Complete Fig. 4.2 to show the mechanism for this reaction.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

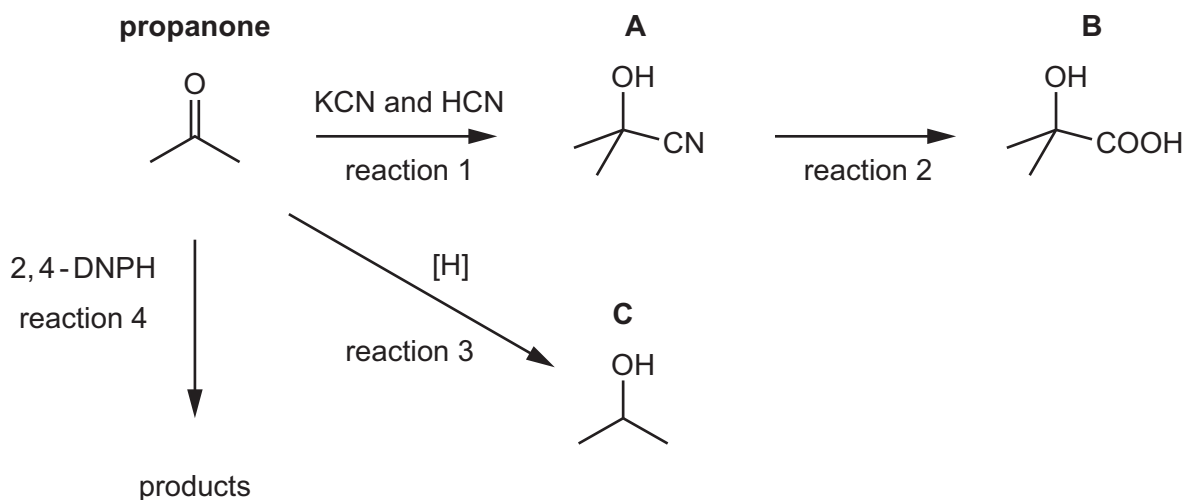
**Fig. 4.2**

[4]

8. O/N 24/P21/Q4(a)(i)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 is a nucleophilic addition reaction.

- (i)** Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.

**Fig. 4.2**

9. O/N 24/P21/Q4(c)

Context

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

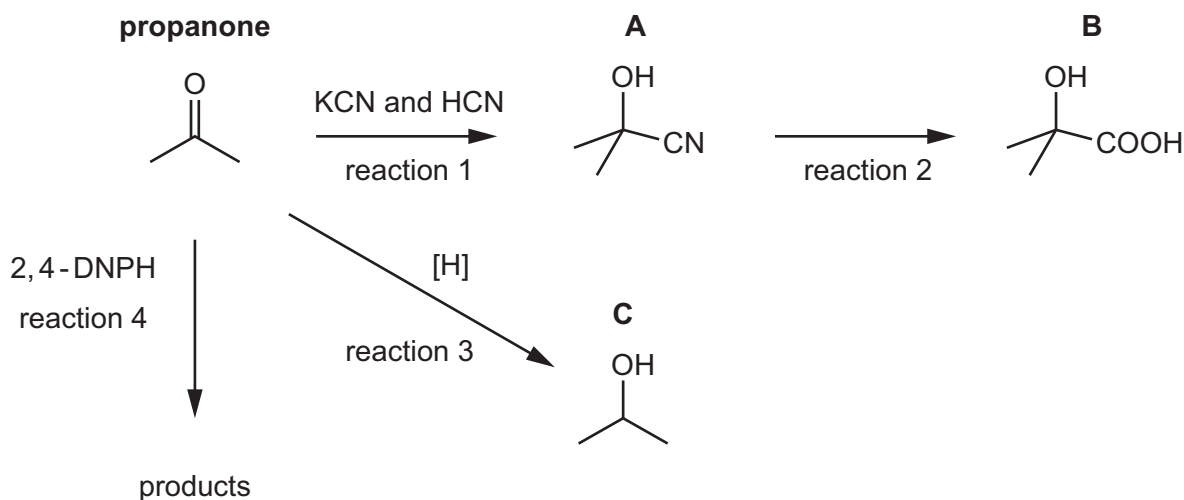


Fig. 4.1

- (c) Reaction 3 is a reduction reaction.

- (i) Construct an equation to represent reaction 3.

Use [H] to represent one atom of hydrogen from the reducing agent.

[1]

- (ii) Name C.

[1]

10. O/N 24/P21/Q4(d)

Context

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

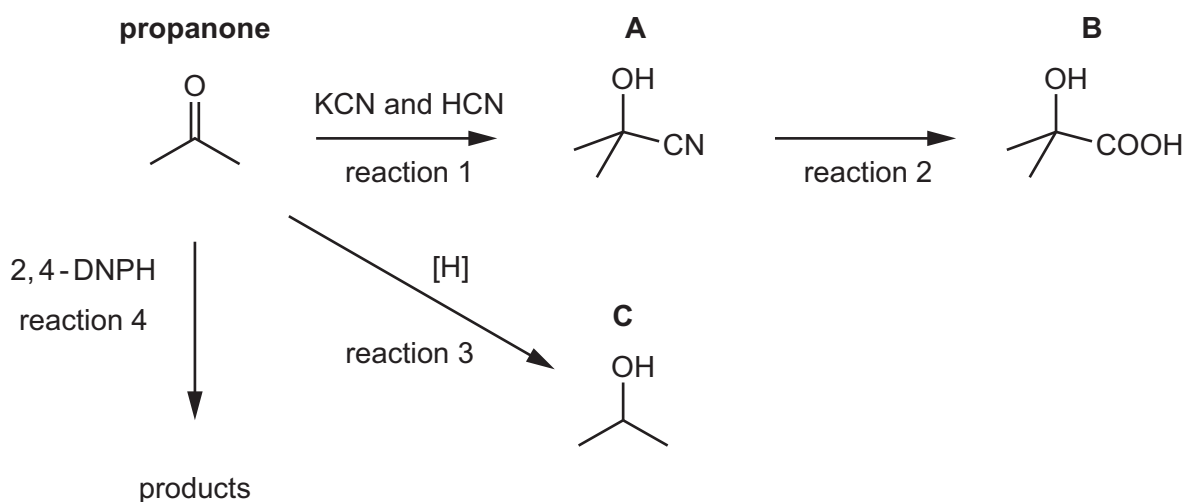


Fig. 4.1

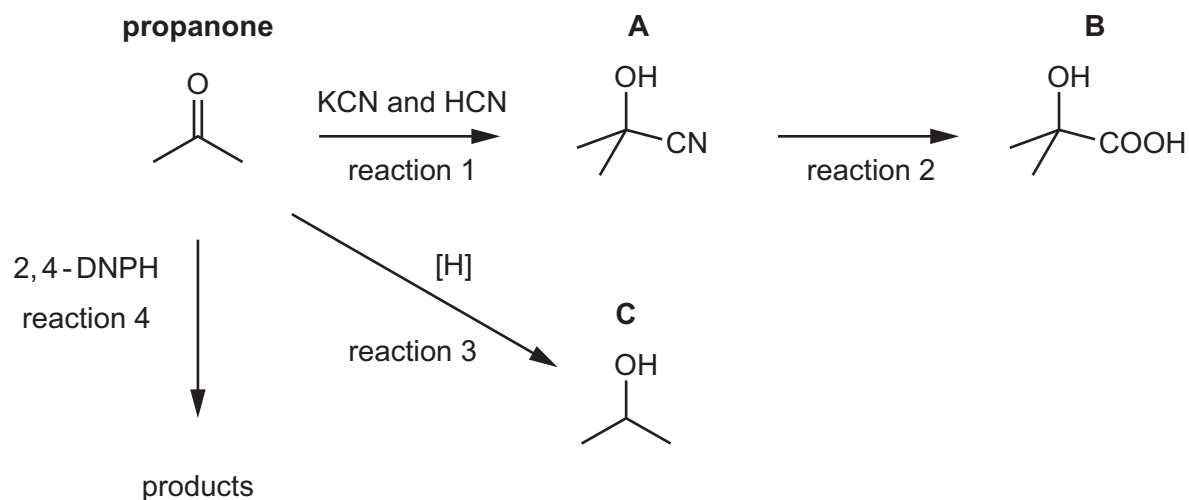
- (d) State what is observed in reaction 4.

[1]

11. O/N 24/P21/Q4(e)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

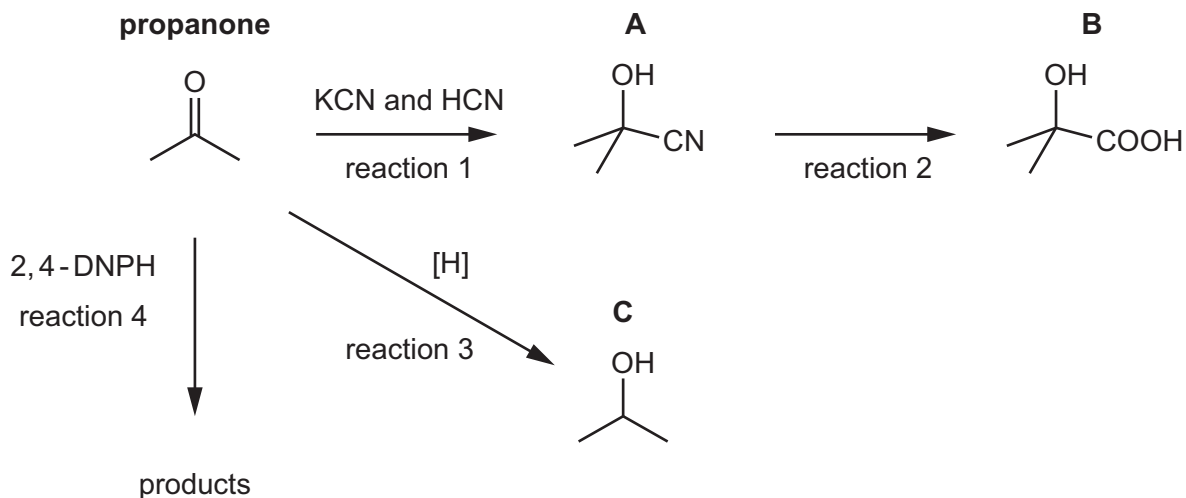
- (e)** Explain why Fehling's reagent does **not** react with propanone.

[1]

12. O/N 24/P23/Q4(a)(i)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 is a nucleophilic addition reaction.

- (i)** Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

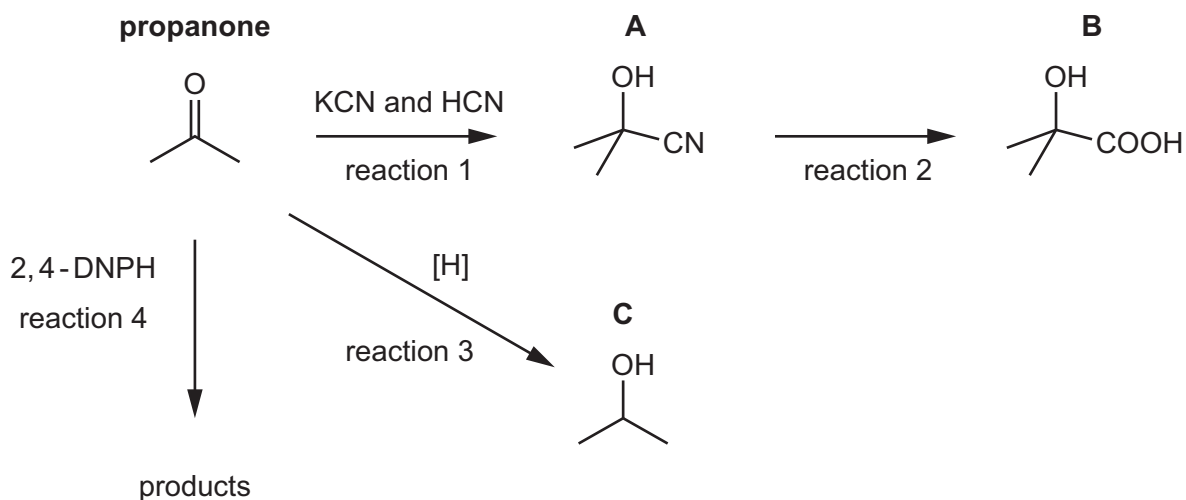
Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.

**Fig. 4.2**

13. O/N 24/P23/Q4(c)

Context

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

- (c) Reaction 3 is a reduction reaction.

- (i) Construct an equation to represent reaction 3.

Use [H] to represent one atom of hydrogen from the reducing agent.

[1]

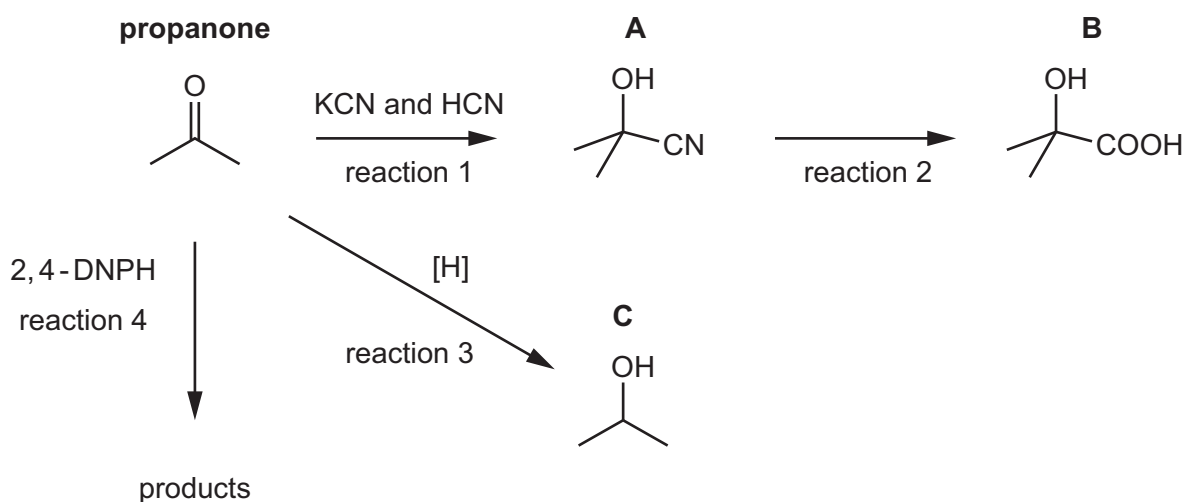
- (ii) Name C.

[1]

14. O/N 24/P23/Q4(d)

Context

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

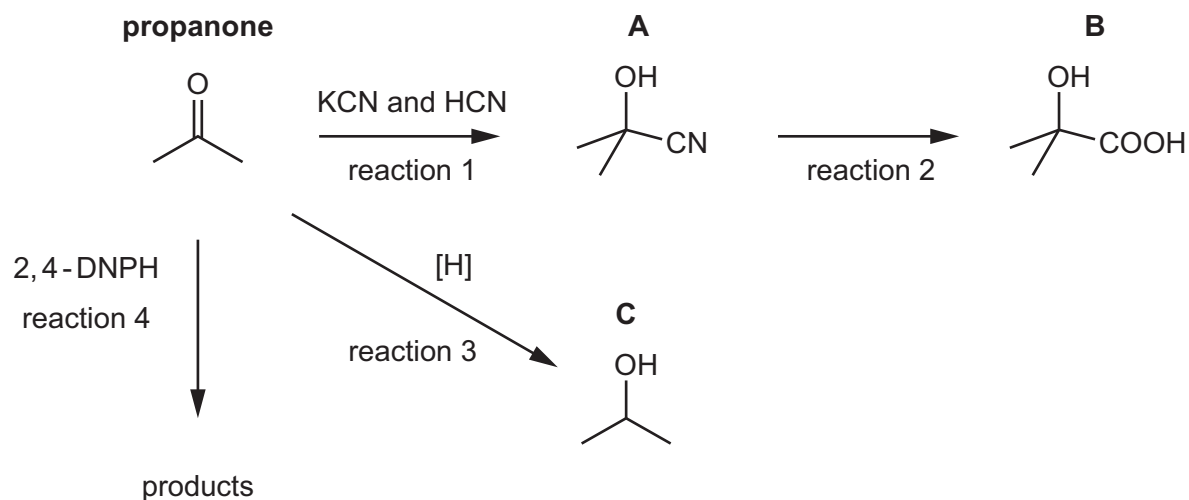
- (d) State what is observed in reaction 4.

[1]

15. O/N 24/P23/Q4(e)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

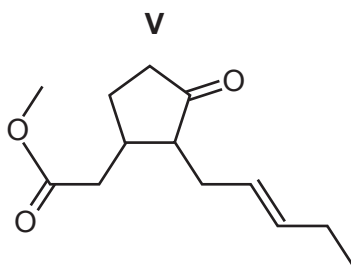
**Fig. 4.1**

- (e)** Explain why Fehling's reagent does **not** react with propanone.

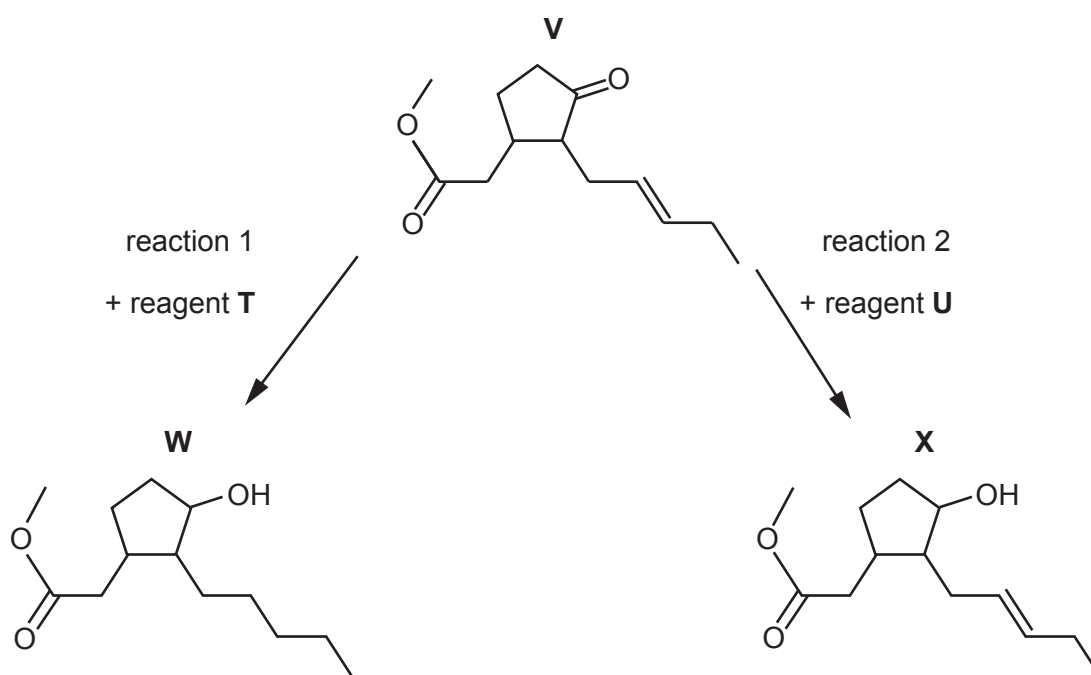
[1]

16. M/J 24/P21/Q6(b)(i)

Context: part (a) (introduction)

6 (a) V shows stereoisomerism.**Fig. 6.1**

Context: part (b) (introduction)

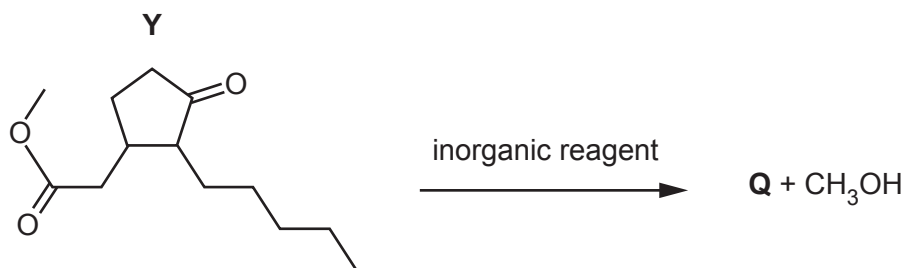
(b) Fig. 6.2 shows two reactions involving **V**.**Fig. 6.2****(i)** Identify the role of reagent **T** for each functional group that reacts in reaction 1.

[1]

17. M/J 24/P21/Q6(c)(ii)

Context: part (c) (introduction)

- (c) Both functional groups in one molecule of **Y** react with an inorganic reagent to form one molecule of **Q** and one molecule of methanol, CH_3OH , as shown in Fig. 6.3.

**Fig. 6.3**

- (ii) **Q** contains **only** hydroxyl functional groups.

Complete Table 6.1 to show the observations that occur when 2,4-dinitrophenylhydrazine (2,4-DNPH reagent) is added to separate samples of **Y** and **Q**.

Table 6.1

	observation on addition of 2,4-DNPH reagent
Y	
Q	

[1]

18. MAR 24/P22/Q1(d)(iii)-(iv)

Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

Context: part (d) (introduction)

- (d)** NaBiO_3 is an oxidising agent with similar properties to KMnO_4 .

Fig. 1.1 shows an example of the use of NaBiO_3 as an oxidising agent.

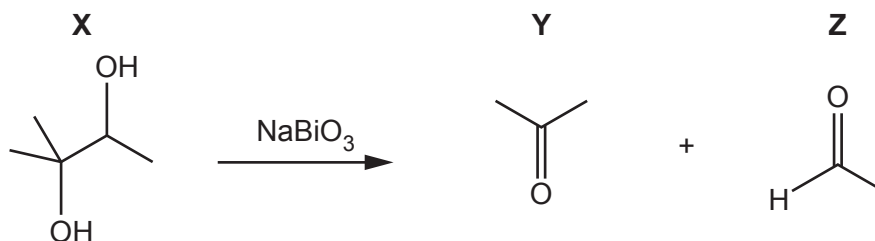


Fig. 1.1

- (iii)** Complete Table 1.1 to show what is observed when compounds **Y** and **Z** react separately with the named reagents.

Table 1.1

reagent	observation with Y	observation with Z
$\text{Na}_2\text{CO}_3(\text{aq})$	no reaction	
alkaline $\text{I}_2(\text{aq})$		
2,4-dinitrophenylhydrazine (2,4-DNPH)		
Tollens' reagent		

[4]

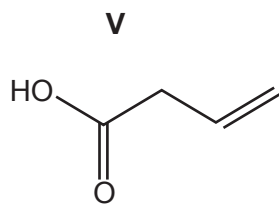
- (iv)** Construct an equation for the reaction of **Z** with NaBH_4 .

Use $[\text{H}]$ to represent an atom of hydrogen from the reducing agent.

[1]

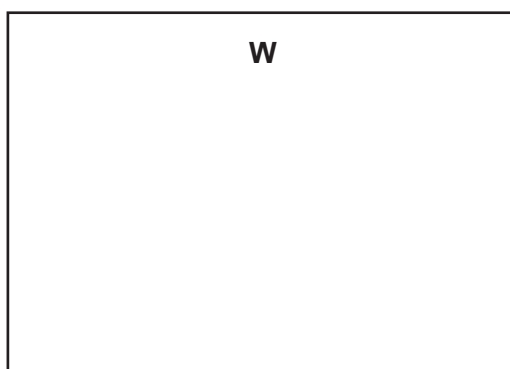
19. M/J 23/P21/Q4(a)

Context

4 **V** is a colourless liquid.**Fig. 4.1**

(a) **V** reacts with an excess of LiAlH_4 to form **W**.

(i) Draw the structure of **W** in the box.



[1]

(ii) Identify the role of LiAlH_4 in the reaction with **V**.

[1]

20. M/J 23/P22/Q5(b)

(b) **A** and **B** are structural isomers with molecular formula $C_5H_{10}O$.

They are both straight-chained molecules with only one functional group.

Table 5.1 describes observations when separate samples of **A** and **B** are added to different reagents.

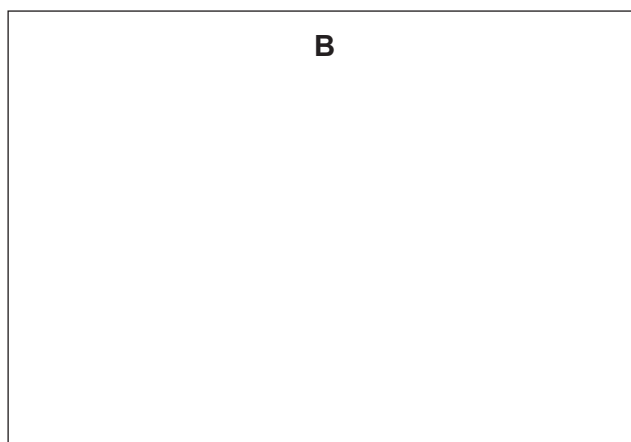
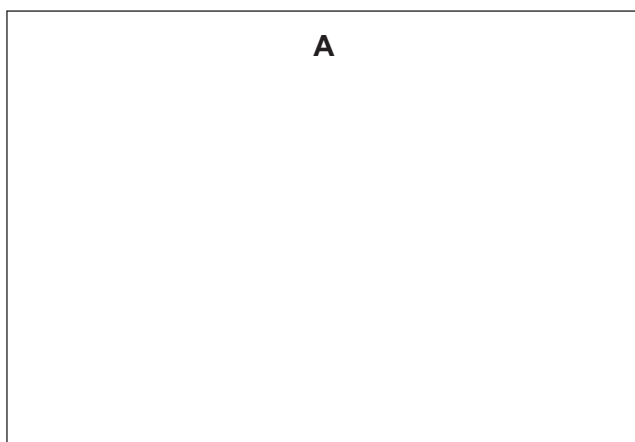
Table 5.1

reagent	A	B
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	orange precipitate appears	orange precipitate appears
Tollens' reagent	silver mirror appears	no reaction
alkaline $I_2(aq)$	no reaction	no reaction

(i) Name the functional group present in both **A** and **B**.

[1]

(ii) Draw the structures of **A** and **B** in the boxes.

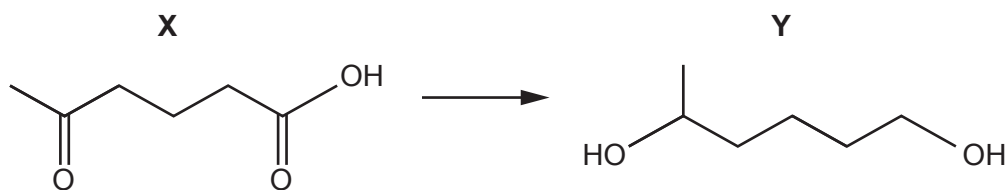


[2]

21. M/J 23/P23/Q5(b)

Context

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.

**Fig. 5.1**

(b) The formation of **Y** from **X** requires the addition of a suitable reducing agent.

- (i)** Construct an equation using molecular formulae and [H] for the reaction in Fig. 5.1. Use [H] to represent one atom of hydrogen from the reducing agent.

[1]

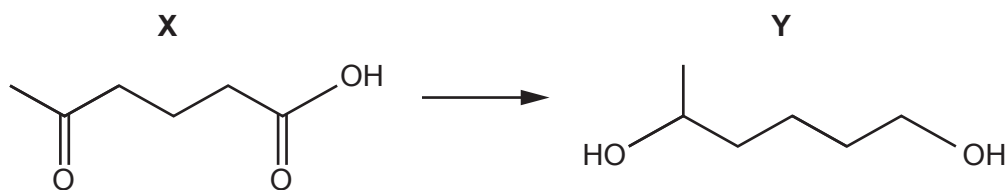
- (ii)** Identify a suitable non-gaseous reducing agent for the formation of **Y** from **X**.

[1]

22. M/J 23/P23/Q5(d)

Context

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.

**Fig. 5.1**

(d) Complete Table 5.2 with the expected observations that occur when the reagents shown are added to separate solutions of **X** and **Y**. Do **not** refer to temperature changes in your answer.

Table 5.2

reagent	observation on addition to X	observation on addition to Y
aqueous sodium carbonate		
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)		
alkaline aqueous iodine		

[3]

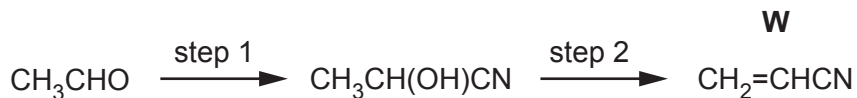
23. M/J 23/P23/Q6(b)(i)-(ii)

Context

- 6** Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

Context: part (b) (introduction)

- (b)** CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN .

**Fig. 6.1**

- (i)** Name the mechanism for the reaction in step 1 in Fig. 6.1.

[1]

- (ii)** Complete Fig. 6.2 to show the mechanism for the reaction in step 1.
Include all products, charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 6.2**

[3]

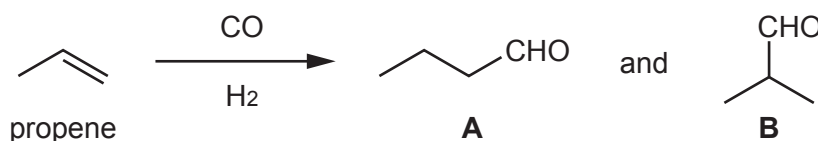
24. MAR 23/P22/Q3(b)

also Unit 13 (1 mark); this unit 8 marks

Context

- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2 .

**Fig. 3.1**

- (b)** **A** and **B** show reactions typical of aliphatic aldehydes.

- (i)** **A** undergoes a nucleophilic addition reaction with a mixture of HCN and KCN , forming compound **C**.

Complete the diagram to show the mechanism for this reaction.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

Draw the structure of the organic intermediate.



[4]

(ii) Table 3.1 shows information about three experiments involving **B**.

Complete Table 3.1.

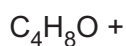
Table 3.1

experiment	reagents	observation with B
1		solution turns from orange to green
2		a silver mirror forms on the sides of the reaction vessel
3	$\text{Br}_2(\text{aq})$	

[3]

(iii) **B**, $\text{C}_4\text{H}_8\text{O}$, is oxidised by acidified potassium manganate(VII).

Complete the equation for this reaction. Use [O] to represent one atom of oxygen from the oxidising agent.



[1]

(iv) **C** is a chiral molecule.

Circle any chiral centres in the structure of **C** shown in Fig. 3.2.

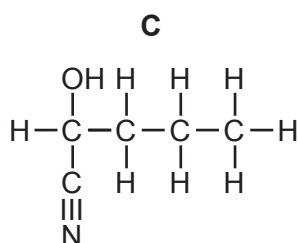


Fig. 3.2

[1]

25. O/N 22/P21/Q5(a)(ii)-(iii)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

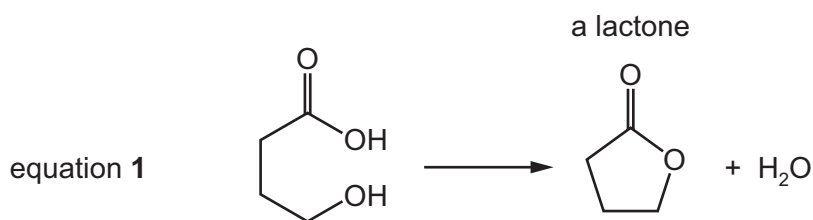
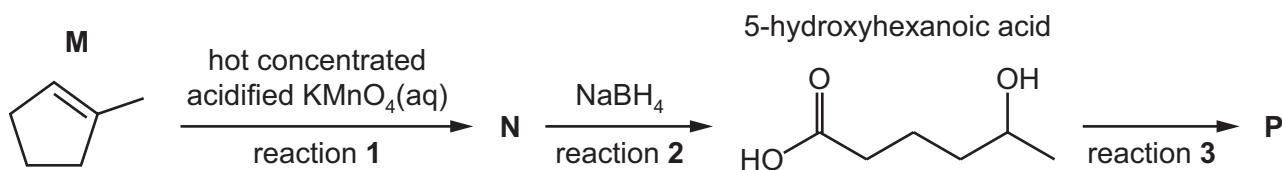


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

**Fig. 5.1**

- (ii) **N** is reduced by NaBH₄ to form 5-hydroxyhexanoic acid in reaction 2.

Construct an equation for reaction 2 using molecular formulae.

In the equation, use [H] to represent one atom of hydrogen from the reducing agent.

[1]

- (iii) Reaction 2 is a nucleophilic addition.

Suggest why reaction 2 creates a mixture of two organic compounds.

[2]

26. O/N 22/P23/Q5(a)(ii)-(iii)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

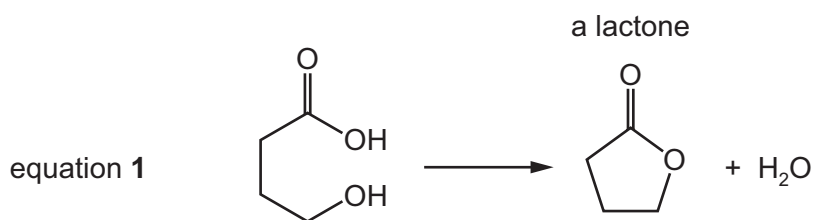
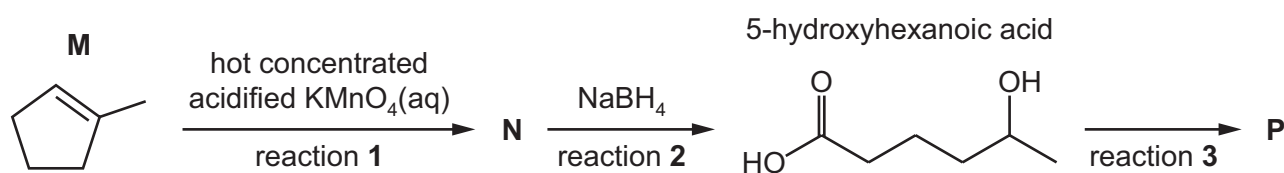


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

**Fig. 5.1**

- (ii)** **N** is reduced by NaBH_4 to form 5-hydroxyhexanoic acid in reaction 2.

Construct an equation for reaction 2 using molecular formulae.

In the equation, use $[\text{H}]$ to represent one atom of hydrogen from the reducing agent.

[1]

- (iii)** Reaction 2 is a nucleophilic addition.

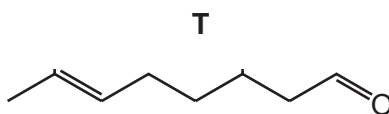
Suggest why reaction 2 creates a mixture of two organic compounds.

[2]

27. M/J 22/P21/Q3(c)(i)

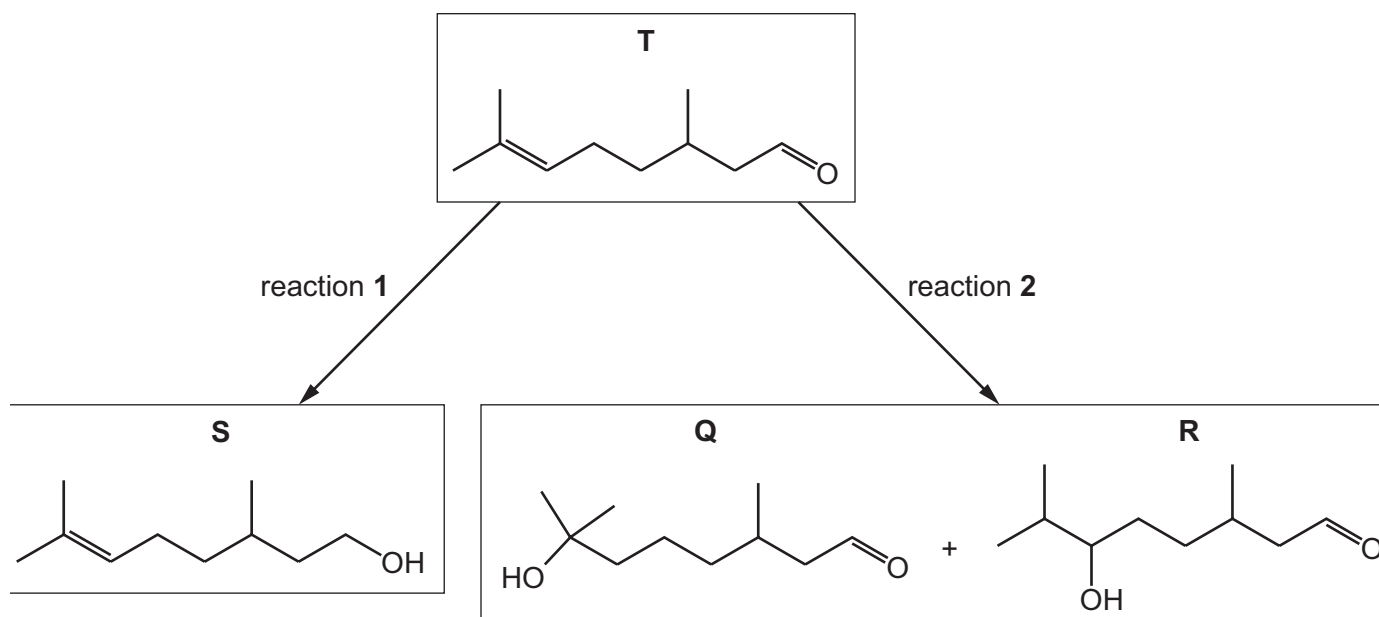
Context

3 Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

Context: part (c) (introduction)

(c) Fig. 3.2 shows two reactions of **T**.

**Fig. 3.2**

(i) Identify a suitable reagent for reaction 1.

[1]

28. M/J 22/P21/Q4(c)(i)-(ii)

Context

4 Compound **V** is a liquid.**V** contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.**V** has a relative molecular mass of 280.

Context: part (c) (introduction)

(c) **W**, **X** and **Y** have the same molecular formula, $C_5H_{10}O$.**W**, **X** and **Y** are added separately to different reagents. Observations for these reactions are described in Table 4.1.**Table 4.1**

	+ 2,4-dinitrophenylhydrazine	+ alkaline $I_2(aq)$	+ Fehling's reagent and warm
W	orange precipitate seen	no change	orange-red precipitate seen
X	orange precipitate seen	yellow precipitate seen	no change
Y	orange precipitate seen		

(i) **W**, **X** and **Y** each contain a common functional group.

Name the functional group that is present in all three compounds.

[1]

(ii) State the formula of the yellow precipitate produced when **X** is added to alkaline $I_2(aq)$.

[1]

29. M/J 22/P23/Q3(c)

Context

3 **G** belongs to a group of compounds called ethers.

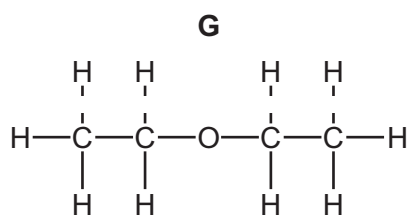


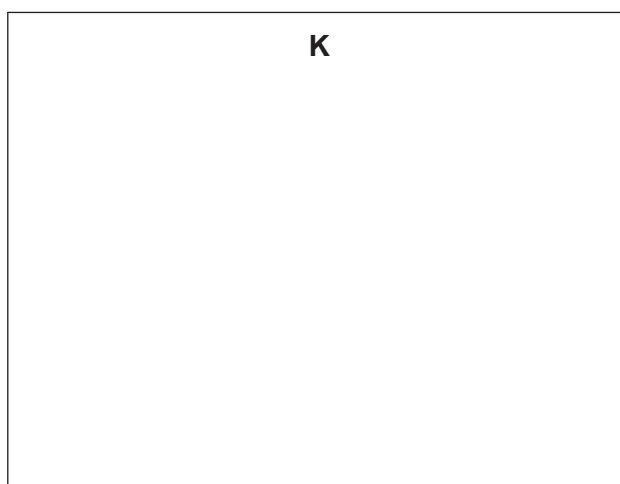
Fig. 3.1

(c) **K** has molecular formula $\text{C}_3\text{H}_6\text{O}$.

When **K** is added to 2,4-dinitrophenylhydrazine, an orange precipitate forms.

When **K** is warmed with Tollens' reagent, a silver mirror forms.

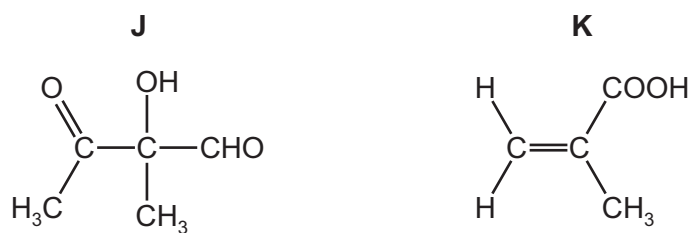
Draw the displayed formula of **K**.



[2]

30. MAR 22/P22/Q4(a)(i)

Context

4 Compounds **J** and **K** are found in plant oils.**Fig. 4.1**

(a) (i) Complete Table 4.1 to state what you would **observe** when **J** reacts with the reagents listed.

Table 4.1

reagent	observation with J
2,4-dinitrophenylhydrazine (2,4-DNPH)	
Tollens' reagent	
sodium metal	

[3]

Answers Section

1. M/J 26/P21/Q4(b)(ii)

4(b)(ii)	Alcohols / Compound A are / is not oxidised by Fehling's reagent (Aldehydes are oxidised by Fehling's reagent) OR Aldehydes are oxidised by Fehling's reagent (Alcohols / Compound A are / is not oxidised by Fehling's reagent)	1 A for alcohol groups = hydroxy = hydroxyl = OH group R for alcohol groups = hydroxide / OH⁻ group A alcohol groups are not a strong enough reducing agent (to reduce Fehling's) A Fehling's reagent is not a strong enough oxidising agent to oxidise the alcohol groups A Fehling's is a strong enough oxidising agent to oxidise aldehydes A Fehling's is too mild to oxidise alcohols A Fehling's isn't reduced by compound A / alcohols A Fehling's is reduced by aldehydes I Fehling's is a mild oxidising agent unqualified I aldehydes are easier to oxidise than alcohols I compound A has no aldehyde functional groups. I reacts with Fehling's I carbonyl R ketones are oxidised by Fehling's
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2. M/J 26/P22/Q6(c)(iii)

6(c)(iii)	M1 Br₂ (aq) / bromine water M2 orange / brown to colourless in S (not R) OR M1 Na₂CO₃ / sodium carbonate / NaHCO₃ / sodium hydrogencarbonate M2 fizzing / effervescence in R (not S)	2 R capitalisation once e.g. KMNO₄ M2 observation dependent on unambiguous attempt at M1 reagent e.g. M1 Br × M2 decolourises S ✓ OR M1 K₂Cr₂O₇ / Cr₂O₇²⁻ (no acid) ^ M2 (orange) to green ✓ OR M1 NaCO₃ × M2 bubbles in R ✓ etc M2 I alkene for S / carboxylic acid for R but A alcohol for S M1 A bromine / Br₂ M1 A aqueous iodine / aqueous I₂ / iodine water NOT iodine alone M1 R iodoform M2 A decolourises M2 A yellow to colourless M2 R red M2 A bubbles M2 I limewater (turns milky)
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6(c)(iii)		Alternatives: M1 acidified potassium dichromate / H^+ AND $\text{K}_2\text{Cr}_2\text{O}_7$ M1 BOD H^+ / $\text{Cr}_2\text{O}_7^{2-}$ M2 (orange) to green with S (not R) M1 potassium permanganate / KMnO_4 M1 BOD potassium manganate M1 BOD MnO_4^- M2 decolorises / (purple) to colourless (solution) / brown ppt with S (not R) M2 A green solution formed if alkaline KMnO_4 used M1 addition of named alcohol and (conc) sulfuric acid OR named alcohol and catalyst M2 sweet smell with R (not S) M1 addition of named carboxylic acid and (conc) sulfuric acid OR named carboxylic acid AND catalyst M2 sweet smell with S (and not R) M1 named acid-base indicator M2 appropriate colour change with R (and not S) M1 A Mg / magnesium M2 A faster bubbles with R (comparison required)
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3. M/J 26/P23/Q4(a)(iii)

4(a)(iii)	Alcohols / Compound V are / is not oxidised by Fehling's reagent (aldehydes are oxidised by Fehling's reagent) OR Aldehydes are oxidised by Fehling's reagent (alcohols / Compound V are / is not oxidised by Fehling's reagent)	1 A for alcohol groups = hydroxy = hydroxyl = OH group R for alcohol groups = hydroxide / OH^- group A alcohol groups are not a strong enough reducing agent (to reduce Fehling's) A Fehling's reagent is not a strong enough oxidising agent to oxidise the alcohol groups A Fehling's is a strong enough oxidising agent to oxidise aldehydes A Fehling's is too mild to oxidise alcohols A Fehling's is not reduced by V / alcohols A Fehling's is reduced by aldehydes I Fehling's is a mild oxidising agent (unqualified) I aldehydes are easier to oxidise than alcohols I compound V has no aldehyde functional groups. I reacts with Fehling's I carbonyl R ketones are oxidised by Fehling's
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4. M/J 25/P21/Q6(a)

6(a)	carbonyl	1
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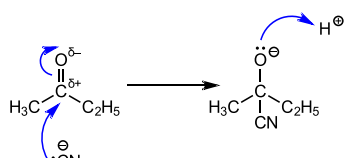
5. M/J 25/P21/Q6(b)

6(b)	M1 (2,4)-dinitrophenylhydrazine / (2,4)-DNPH M2 orange / red / yellow precipitate	2
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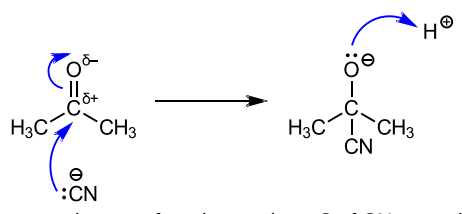
6. M/J 25/P21/Q6(c)(i)-(ii)

6(c)(i)	F = propanone G = ethanal H = propanal	2
6(c)(ii)	tri-iodomethane OR CHI_3	1

7. MAR 25/P22/Q4(b)(i)

4(b)(i)	 • curly arrow from lone pair on :CN^- to $\text{C}(\delta^+)$ • curly arrow from $\text{C}=\text{O}$ to $\text{O}(\delta^-)$ AND correct dipole • correct intermediate • curly arrow from lone pair on O^- to H^+	4
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8. O/N 24/P21/Q4(a)(i)

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on $\text{C}=\text{O}$ • curly arrow from $\text{C}=\text{O}$ bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O^- to (source of) H^+	1
		1
		1

9. O/N 24/P21/Q4(c)

4(c)(i)	$\text{CH}_3\text{COCH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	1
4(c)(ii)	propan-2-ol	1

10. O/N 24/P21/Q4(d)

4(d)	red / orange / yellow ppt	1
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11. O/N 24/P21/Q4(e)

4(e)	Fehling's solution cannot oxidise ketones OR ketones are not easily oxidised	1
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12. O/N 24/P23/Q4(a)(i)

4(a)(i)	• curly arrow from lone pair on C of CN⁻ to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O⁻ to (source of) H⁺	1
		1
		1

13. O/N 24/P23/Q4(c)

4(c)(i)	$\text{CH}_3\text{COCH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	1
4(c)(ii)	propan-2-ol	1

14. O/N 24/P23/Q4(d)

4(d)	red / orange / yellow ppt	1
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15. O/N 24/P23/Q4(e)

4(e)	Fehling's solution cannot oxidise ketones OR ketones are not easily oxidised	1
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16. M/J 24/P21/Q6(b)(i)

6(b)(i)	reducing agent for C=O AND C=C	1
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17. M/J 24/P21/Q6(c)(ii)

6(c)(ii)		observation on addition of 2,4-DNPH	1
	Y	orange precipitate	
	Q	no precipitate	
	Both correct for one mark		

18. MAR 24/P22/Q1(d)(iii)-(iv)

1(d)(iii)	<table border="1"> <tr> <th>reagent</th><th>observation with Y</th><th>observation with Z</th></tr> <tr> <td>$\text{Na}_2\text{CO}_3(\text{aq})$</td><td>no reaction</td><td>• no reaction</td></tr> <tr> <td>alkaline $\text{I}_2(\text{aq})$</td><td>• yellow precipitate (ppt)</td><td>• yellow ppt</td></tr> <tr> <td>2,4-DNPH</td><td>• red / orange / yellow ppt</td><td>• red / orange / yellow ppt</td></tr> <tr> <td>Tollens' reagent</td><td>• no reaction</td><td>• silver mirror</td></tr> </table>	reagent	observation with Y	observation with Z	$\text{Na}_2\text{CO}_3(\text{aq})$	no reaction	• no reaction	alkaline $\text{I}_2(\text{aq})$	• yellow precipitate (ppt)	• yellow ppt	2,4-DNPH	• red / orange / yellow ppt	• red / orange / yellow ppt	Tollens' reagent	• no reaction	• silver mirror	4
reagent	observation with Y	observation with Z															
$\text{Na}_2\text{CO}_3(\text{aq})$	no reaction	• no reaction															
alkaline $\text{I}_2(\text{aq})$	• yellow precipitate (ppt)	• yellow ppt															
2,4-DNPH	• red / orange / yellow ppt	• red / orange / yellow ppt															
Tollens' reagent	• no reaction	• silver mirror															
1(d)(iv)	$\text{CH}_3\text{CHO} + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{OH}$	1															

19. M/J 23/P21/Q4(a)

4(a)(i)		1
4(a)(ii)	reducing agent	1

20. M/J 23/P22/Q5(b)

5(b)(i)	carbonyl (group)	1
5(b)(ii)		2

21. M/J 23/P23/Q5(b)

5(b)(i)	$\text{C}_6\text{H}_{10}\text{O}_3 + [6\text{H}] \rightarrow \text{C}_6\text{H}_{14}\text{O}_2 + \text{H}_2\text{O}$	1
5(b)(ii)	LiAlH_4 (in dry ether)	1

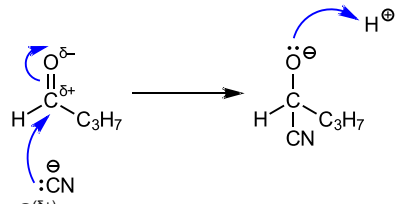
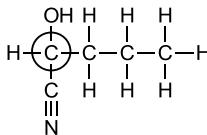
22. M/J 23/P23/Q5(d)

5(d)	reagent	observation on addition to X	observation on addition to Y	3
	sodium carbonate solution	effervescence	no change	
	2,4-DNPH	orange precipitate	no change	
	alkaline aqueous iodine	yellow precipitate	yellow precipitate	
	Two correct for one mark, four correct for two marks, six correct for three marks			

23. M/J 23/P23/Q6(b)(i)-(ii)

6(b)(i)	nucleophilic addition	1
6(b)(ii)	M1 correct dipole on C=O AND curly arrow from C=O bond to O AND from lone pair of electrons on C of :CN⁻ M2 correct intermediate (charge on O or overall) M3 curly arrow from lone pair of electrons on O to H⁺ AND correct organic product (+ CN⁻)	3

24. MAR 23/P22/Q3(b)

3(b)(i)	 curly arrow from lone pair on C of CN⁻ to C(^{δ+}) correct dipole on C=O AND curly arrow from C=O bond to O^{δ-} correct organic intermediate curly arrow from lone pair on alkoxide O⁽⁻⁾ to H⁺ / H of HCN / H of H₂O (to reform the alcohol)	1
3(b)(ii)	acidified K ₂ Cr ₂ O ₇	1
	Tollens' reagent	1
	no reaction	1
3(b)(iii)	C ₄ H ₈ O + [O] → C ₄ H ₈ O ₂	1
3(b)(iv)		1

25. O/N 22/P21/Q5(a)(ii)-(iii)

5(a)(ii)	C ₆ H ₁₀ O ₃ + 2[H] → C ₆ H ₁₂ O ₃	1
5(a)(iii)	M1 (ketone in) N is planar (so can be attacked from either side)	1
	M2 because different stereoisomers / optical isomers form	1

26. O/N 22/P23/Q5(a)(ii)-(iii)

5(a)(ii)	C ₆ H ₁₀ O ₃ + 2[H] → C ₆ H ₁₂ O ₃	1
5(a)(iii)	M1 (ketone in) N is planar (so can be attacked from either side)	1
	M2 because different stereoisomers / optical isomers form	1

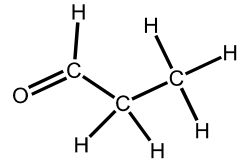
27. M/J 22/P21/Q3(c)(i)

3(c)(i)	NaBH ₄ / sodium borohydride / LiAlH ₄ / lithium tetrahydridoaluminate	1
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28. M/J 22/P21/Q4(c)(i)-(ii)

4(c)(i)	carbonyl	1
4(c)(ii)	CHI ₃	1

29. M/J 22/P23/Q3(c)

3(c)	displayed formula of propanal  M1 identification of propanal M2 correct use of displayed formula and correct formula used	2
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30. MAR 22/P22/Q4(a)(i)

4(a)(i)	red / orange / yellow precipitate / ppt / solid [1] silver mirror / silver / grey solid / precipitate / ppt [1] effervescence / bubbling / fizzing [1]	3
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